

Olympiad Test Paper — Social Science (Class 7) — Paper 5

Chapter 2: Understanding the Weather

Paper 5 — (progressively harder)

Subject	Social Science (NCERT Class 7)
Chapter	2 — Understanding the Weather
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: confusing weather with climate, assuming pressure rises with altitude (it falls), swapping the jobs of thermometer/barometer/rain gauge/wind vane/anemometer/hygrometer, treating high humidity as a guarantee of rain, and mixing up the units mb, degrees, percent and mm; many stems need multi-step reasoning, data reading or a full cause-and-effect chain. Do not write on this paper — mark answers on your answer sheet.

1. A diary keeps two records of Darjeeling: a line for 12 March, '6 degrees, foggy, light drizzle since noon', and a paragraph beginning 'Decade after decade, the spring stays cool and misty here.' A student must label each correctly. The right labels are:

- (A) The dated 12 March line is climate because it gives an exact temperature, and the decade paragraph is weather because it has no number
- (B) The dated 12 March line is weather (the state of the air at one time) and the decade-after-decade paragraph is climate (the long-term average pattern)
- (C) Both records are weather because both name Darjeeling and mention mist
- (D) Both records are climate because both describe cool, misty conditions

2. Mawsynram is famous for a very wet climate, yet a tourist there on one April afternoon enjoyed clear blue skies. A friend concludes, 'Then Mawsynram's climate must be dry now.' The soundest reply is:

- (A) The friend is right, because one clear afternoon is enough to redefine the climate
- (B) Clear skies are impossible anywhere with a wet climate, so the tourist must be mistaken

(C) One dry afternoon is just that day's weather; the climate, being the average over many years, is unchanged by a single day

(D) Weather and climate mean the same thing, so the climate has indeed turned dry

3. Two coastal towns sit at almost the same latitude and so receive sunlight for similar lengths of time, yet one has a far wetter climate than the other. A pupil insists this is impossible. The best correction is:

(A) Climate is shaped by more than day-length, so two places at one latitude can still settle into different long-term patterns

(B) It is indeed impossible, because equal sunlight hours force an identical climate

(C) The wetter town must be noticeably closer to the Sun than the drier one

(D) Climate is fixed only by how many weather instruments each town owns

4. A farmer says, 'For deciding whether to harvest in the next hour I need one kind of information; for choosing which orchard trees to plant for the next twenty years I need another.' The correct pairing is:

(A) The next-hour harvest decision needs the climate, while the twenty-year planting choice needs the weather

(B) The next-hour harvest decision needs the weather, while the twenty-year planting choice needs the climate

(C) Both decisions depend only on the weather of the coming hour

(D) Both decisions depend only on the thirty-year climate average

5. A report states: 'July in this town has been hot and humid in most years on record; today it is unusually cool with a sharp shower.' Which single conclusion is justified?

(A) Today's coolness proves the climate has permanently become cool and rainy

(B) The 'most years' statement is the climate and 'today' is the weather, so today's coolness is an exception that does not change the climate

(C) Both halves of the report describe the climate because both mention July

(D) Both halves describe the weather because both describe real conditions

6. A class debates: 'A place with a hot, dry desert climate can never have a cold or rainy day.' Why is this wrong?

(A) It is correct: a desert climate forbids any cold or rainy day at all

(B) It is wrong only because deserts actually have wet climates

(C) Climate is only the long-term average, so the day-to-day weather can swing away from it, including the odd cold or rainy day

(D) It is wrong because weather and climate are identical, so the climate itself rains some days

7. Nearly all clouds, rain and storms occur in the troposphere. Using only chapter ideas, the best reason this lowest layer is where weather happens is that:

- (A) It is the highest, thinnest layer, farthest from the Earth's surface
- (B) It touches the ground, where the Sun-warmed surface, water vapour and rising air can all interact
- (C) It lies in empty space beyond the reach of the Sun's heat
- (D) It is the layer in which the oceans store most of their water

8. The troposphere reaches higher over the warm tropics than over the cold poles. Which cause-and-effect chain matches the chapter?

- (A) Cold polar air expands more than warm tropical air, lifting the layer higher at the poles
- (B) The tropics sit physically much nearer the Sun, which pulls the layer outward
- (C) Heavier rainfall at the poles thickens the layer there
- (D) Warm air expands and rises, pushing the layer higher over the tropics, while cold air contracts and keeps the layer lower over the poles

9. A child says, 'The atmosphere is just empty sky, so it cannot weigh anything or press on us.' According to the chapter, the correct reply is:

- (A) The atmosphere is solid rock, which is why it has weight
- (B) The atmosphere is a blanket of gases with real weight, and that weight pressing down is what we call atmospheric pressure
- (C) The atmosphere is made only of water, which is why it falls as rain
- (D) The atmosphere has no weight, yet it still causes wind for no clear reason

10. A mountaineer leaves a coast where the barometer reads 1012 mb and climbs to a high pass. Which single statement correctly links her height, the pressure and her breathing?

- (A) As she climbs, she nears the top of the sky, so the pressure rises and breathing gets easier
- (B) The pressure stays exactly 1012 mb the whole way up, so her breathing never changes
- (C) The pressure leaps to zero at the pass, so there is no air at all to breathe
- (D) As she climbs, less air is left above her, so the pressure falls and the thinner air gives less oxygen per breath

11. At the same hour, station P reads 1010 mb, station Q reads 870 mb and station R reads 700 mb, and none of the barometers is faulty. Ordered from the LOWEST place to the HIGHEST, the stations are:

- (A) R lowest, then Q, then P highest, because the largest pressure number sits highest up
- (B) Q lowest, then P, then R highest, because pressure changes at random with height
- (C) All three are at the same height, because air spreads everywhere equally
- (D) P lowest, then Q, then R highest, because pressure falls as height increases

12. Assertion: A barometer carried up a mountain shows a smaller reading at the top than at the foot. Reason: There is less air above the top of the mountain to press down than above its foot. Choose the best option.

- (A) Both the assertion and the reason are true, and the reason correctly explains the assertion
- (B) The assertion is true but the reason is false
- (C) The assertion is false but the reason is true
- (D) Both are true, but the reason does not explain the assertion

13. A weather club sorts cards into 'element of weather' and 'land or water feature'. Which set of four cards belongs ENTIRELY in the 'element of weather' pile?

- (A) Temperature, atmospheric pressure, humidity, precipitation
- (B) Temperature, rivers, wind, humidity
- (C) Pressure, mountains, rainfall, sunshine
- (D) Wind, soil type, temperature, precipitation

14. Three readings are stripped of their units: 1009, 28, 64, belonging to pressure, temperature and humidity. The only sensible assignment is:

- (A) 1009 is the temperature in degrees, 28 is the pressure in millibars, 64 is the humidity in percent
- (B) 1009 is the humidity in percent, 28 is the temperature in degrees, 64 is the pressure in millibars
- (C) 1009 is the pressure in millibars, 28 is the humidity in percent, 64 is the temperature in degrees
- (D) 1009 is the pressure in millibars, 28 is the temperature in degrees, 64 is the humidity in percent

15. A pupil wants to add 'the time of sunrise' to the list of weather elements. The best reason to reject this is:

- (A) Sunrise time is set by the motion of Earth and Sun, not by the changing state of the air, so it is not a weather element
- (B) Sunrise time changes far too quickly to count as a weather element
- (C) Sunrise time is really a form of precipitation
- (D) Sunrise time is measured by a barometer, so it is just pressure

16. Why do weather reports give each element a precise number on a shared scale rather than words such as 'warm' or 'breezy'?

- (A) Numbers make the weather itself change more slowly
- (B) A measured value on a fixed scale means the same to everyone everywhere, so readings can be compared and combined
- (C) Words such as 'warm' are forbidden in any weather report
- (D) Only the instruments, never people, can read the numbers

17. In a liquid-in-glass thermometer the coloured column climbs as the day warms. The correct cause-and-effect chain is:

- (A) Heat widens the glass tube, letting the liquid spread upward
- (B) Heat makes the liquid expand, and with room to grow only up the narrow tube, the level rises
- (C) Warm air is pushed down into the tube, lifting the liquid
- (D) Heat makes the liquid lighter, so it simply floats out of the bulb

18. A log notes 20 degrees Celsius on Monday and 68 degrees Fahrenheit on Tuesday, and a pupil panics that the thermometer is broken because the numbers are so far apart. The correct reading is:

- (A) 68 Fahrenheit must be far hotter than 20 Celsius because 68 is the bigger number
- (B) Fahrenheit really measures pressure, so the two readings cannot be compared
- (C) The thermometer is faulty, since one temperature must always give one number
- (D) Celsius and Fahrenheit are two scales for the same warmth, so different numbers are expected and the thermometer is fine

19. A station logs a maximum of 39 degrees and a minimum of 21 degrees in one day. Which single statement about its range AND mean is correct?

- (A) The range is 18 degrees and the mean is 18 degrees
- (B) The range is 60 degrees and the mean is 30 degrees
- (C) The range is 18 degrees and the mean is 30 degrees
- (D) The range is 30 degrees and the mean is 18 degrees

20. Five towns reported one day. Town W: max 33, min 27. Town X: max 45, min 18. Town Y: max 29, min 16. Town Z: max 40, min 34. Which town had the LARGEST daily temperature range?

- (A) Town W, because its minimum is fairly high
- (B) Town Z, because it has the second-highest maximum
- (C) Town X, with a range of 27 degrees
- (D) Town Y, because it has the lowest maximum

21. Using the same five towns (W: 33/27, X: 45/18, Y: 29/16, Z: 40/34), which town stayed MOST STEADY across the day, that is had the smallest range?

- (A) Town X, because its numbers are the largest
- (B) Town Y, because it has the lowest minimum
- (C) Town Z alone, because it has the highest minimum
- (D) Town W and Town Z are tied, each with a range of only 6 degrees

22. Two cities share the same mean daily temperature of 28 degrees, but city P has a range of 4 degrees and city Q a range of 22 degrees. What does this most likely tell us?

- (A) City Q must have been hotter than city P at every hour of the day
- (B) City Q had a far greater gap between its day and night temperatures than city P, even though their averages match
- (C) City P must have received less rainfall than city Q
- (D) Two cities cannot share a mean if their ranges differ

23. City M reports max 30, min 20; city N reports max 26, min 24. Comparing them, which statement is correct?

- (A) M and N share the same mean of 25 degrees, but M has the larger range
- (B) M is warmer than N at every hour because its maximum is higher
- (C) N has both the larger mean and the larger range
- (D) M and N have the same range but different means

24. Precipitation is best defined, taking the chapter's full meaning, as:

- (A) Only the rain that falls during the monsoon season
- (B) The invisible water vapour held in the air
- (C) The weight of all the air pressing down on the ground
- (D) Any form of water — rain, snow, sleet or hail — that falls from the sky to the ground

25. Hail and dew are both linked to water, yet differ sharply. Which contrast is correct?

- (A) Both hail and dew fall from the clouds as solid ice
- (B) Hail is invisible vapour, while dew is a strong wind
- (C) Hail falls from the sky as hard balls of ice, while dew forms as droplets settling on cool surfaces without falling from clouds
- (D) Hail is measured in percent, while dew is measured in millibars

26. A rain gauge reports '30 mm' for the day. Two gauges in the same small yard read 30 mm and 4 mm. What is the soundest conclusion?

- (A) 30 mm means the rain would lie 30 mm deep on flat ground, and the 4 mm gauge is probably blocked or faulty since both should read nearly the same
- (B) 30 mm means it rained for exactly 30 minutes, and both gauges are working fine
- (C) 30 mm is a volume of 30 millilitres, and the storm simply missed one gauge
- (D) Rainfall naturally drops to a tenth within a metre, so both readings are correct

27. Snow falls into an open rain gauge overnight and is still frozen at dawn. What is the correct step before reading the scale, and why?

- (A) Read the depth of the solid snow directly, because snow and its meltwater take the same depth
- (B) Let the snow melt to water first, then read the depth, because a rain gauge measures the depth of liquid water
- (C) Tip the snow out and record zero, because frozen water does not count as precipitation
- (D) Double the snow depth, because melting always halves the volume

28. Over one storm, gauge 1 collects water that would lie 18 mm deep and gauge 2 collects water that would lie 9 mm deep, both in the same garden. The most reasonable thought is:

- (A) Rain always falls exactly half as deep every metre, so both readings are right
- (B) Gauge 2 must be measuring the temperature instead of the rain
- (C) Such a large gap over one small garden suggests one gauge is partly blocked or wrongly placed, not that the rain truly halved a few steps away
- (D) The deeper reading proves it rained for twice as long over gauge 1

29. Hail, snow and sleet are sorted as 'precipitation' and humidity as 'not precipitation'. Why does humidity belong in the 'not precipitation' group?

- (A) Humidity is a kind of falling ice, so it should really be grouped with hail
- (B) Humidity is measured in millimetres, just like rainfall, so it is precipitation
- (C) Humidity is the weight of the air, so it is grouped with pressure-driven rain
- (D) Humidity is water vapour still held in the air, whereas precipitation is water that has fallen out of the sky

30. Region G is at 1024 mb and region H beside it is at 996 mb; a second pair, region J at 1012 mb and region K at 1006 mb, lies the same distance apart. Comparing the two pairs, the wind will be:

- (A) Stronger between J and K, because their smaller difference drives a faster wind
- (B) Equal in both pairs, because wind speed does not depend on the pressure difference
- (C) Blowing from H to G and from K to J, that is from low pressure toward high
- (D) Stronger between G and H, because their larger pressure difference drives a faster wind

31. Region M is at 1018 mb and region N beside it at 1002 mb. Which single statement about the wind between them is correct?

- (A) The wind blows from N toward M, because air moves from lower pressure to higher
- (B) The wind spins in a closed circle between them and goes nowhere
- (C) The wind rises straight upward and does not move sideways at all
- (D) The wind blows from M toward N, because air moves from higher pressure to lower pressure

32. Assertion: Wind blows from a low-pressure area toward a high-pressure area. Reason: Air moves so as to even out differences in pressure. Choose the best option.

- (A) The assertion is false but the reason is true
- (B) Both the assertion and the reason are true, and the reason explains the assertion
- (C) Both are true, but the reason does not explain the assertion
- (D) The assertion is true but the reason is false

33. A weather station must record BOTH the direction the wind comes from AND how fast it blows. Which pairing of instruments to jobs is entirely correct?

- (A) An anemometer for the direction, and a wind vane for the speed
- (B) A barometer for the direction, and a thermometer for the speed
- (C) A rain gauge for the direction, and a hygrometer for the speed
- (D) A wind vane for the direction the wind comes from, and an anemometer for the wind's speed

34. At an airport a fabric wind sock hangs limp at dawn but by midday stretches out straight and level. What does the pilot correctly read from this change?

- (A) The wind strengthened from near-calm at dawn to fairly strong by midday
- (B) The wind weakened from strong to calm, since a straight sock means little wind
- (C) The air pressure over the runway rose through the morning
- (D) Heavy rainfall began over the runway by midday

35. At one moment an anemometer's cups barely turn while a wind vane's arrow swings firmly to point north-west. The correct combined reading is:

- (A) The wind is very strong, and the vane is showing its high speed
- (B) The wind is gentle, and the slowly turning cups are showing its direction
- (C) The wind is gentle (low speed) and is coming from the north-west
- (D) The air pressure is high, which is why the cups turn slowly

36. A weather diary records these events one stormy afternoon, but jumbled: dark clouds gathered; the barometer began to fall; rain began; the wind picked up. Which order keeps the cause-and-effect sequence from earliest sign to final result?

- (A) Rain first, then gathering cloud, then strengthening wind, then falling pressure
- (B) Strengthening wind first, then rain, then falling pressure, then gathering cloud
- (C) Falling pressure first, then strengthening wind, then gathering cloud, then rain
- (D) Gathering cloud first, then falling pressure, then rain, then strengthening wind

37. Identify the ODD ONE OUT: three of these instruments report something about moving or pressing air, while one reports water that has fallen. The odd one out is:

- (A) Barometer
- (B) Wind vane
- (C) Anemometer
- (D) Rain gauge

38. A station owns a thermometer, a barometer, a wind vane and an anemometer, but no hygrometer and no rain gauge. Which TWO weather elements can it NOT record?

- (A) Temperature and pressure
- (B) Wind direction and wind speed
- (C) Humidity and rainfall
- (D) Pressure and wind speed

39. Humidity refers to which of the following, and is reported in which unit?

- (A) The weight of the air above us, given in millibars
- (B) The amount of water vapour in the air, given as a percentage
- (C) The depth of rain collected, given in millimetres
- (D) The speed of the wind, given in degrees

40. A hygrometer reads 92 percent at noon and 28 percent by mid-afternoon. Which statement about the air at those two times is correct?

- (A) At noon the air was completely dry, and by afternoon it became nearly saturated
- (B) At noon the air was very damp, nearly as full of vapour as it could hold; by afternoon it was dry and far from full
- (C) At noon it was certainly raining, and by afternoon the temperature reached 28 degrees
- (D) Both readings mean the air held exactly the same amount of vapour

41. Two days share the same temperature. On Monday the relative humidity is 25 percent, on Tuesday it is 88 percent. On which day will wet washing dry FASTER, and why?

- (A) Monday, because the drier air is far from full and can readily absorb more moisture from the clothes
- (B) Tuesday, because the moist air pulls water out of the clothes faster
- (C) Monday, because high pressure on that day squeezes water from the cloth
- (D) Both days equally, because humidity has no effect on drying

42. On a muggy day a hygrometer reads 94 percent, yet no rain falls all afternoon. Which explanation fits the chapter best?

- (A) High humidity alone is not enough; the air must also cool enough for the vapour to condense before rain can form
- (B) High humidity always brings rain, so the hygrometer must be faulty
- (C) Humidity and rainfall have nothing to do with each other
- (D) A reading of 94 percent actually means the air is extremely dry

43. Two towns share the same air temperature. Town A reports 35 percent humidity and town B reports 80 percent. Which single statement is correct?

- (A) Town B's air is drier, so puddles will dry up faster there
- (B) Both towns must be having rain because both have measurable humidity
- (C) Town A's air is drier and can hold much more extra vapour than town B's, so evaporation is easier in town A
- (D) The humidity figures show town A is hotter than town B

44. A pupil claims, 'A barometer and a hygrometer measure pretty much the same thing.' The correct reply is:

- (A) Yes, both of them measure the air pressure
- (B) Yes, both of them measure the humidity
- (C) No — a barometer measures atmospheric pressure while a hygrometer measures humidity, two different elements
- (D) No — a barometer measures rainfall while a hygrometer measures wind speed

45. Putting the instruments together, which single element-to-instrument pairing is INCORRECT?

- (A) Temperature is measured by the thermometer
- (B) Rainfall is measured by the rain gauge
- (C) Wind speed is measured by the anemometer
- (D) Humidity is measured by the barometer

46. Day and night, and the whole pattern of weather, are ultimately driven by which single cause?

- (A) The Moon's gravity pulling on the air
- (B) The spinning of wind vanes at weather stations
- (C) The barometers raising and lowering the pressure
- (D) The Sun's heat, which warms the land, sea and air unevenly

47. Under one midday Sun a bare sandy patch warms much more than a nearby pond, the air over the sand rises, and a breeze flows from the pond toward the sand. Which chain explains how the Sun causes this wind?

- (A) The Sun heats every surface exactly equally, so the air is pushed about at random
- (B) The Sun's rays physically blow the air sideways across the ground
- (C) The Sun heats the two surfaces unevenly, creating a pressure difference, and air flows from the higher to the lower pressure
- (D) The Sun cools the sand, freezing the air so it cannot move

48. Why are the tropics generally warmer than the polar regions, given that all parts of Earth lie at about the same distance from the Sun?

- (A) Sunlight strikes the tropics nearly overhead so its heat is concentrated, while it hits the poles at a slant and spreads thinly
- (B) The tropics are physically much closer to the Sun than the poles
- (C) The poles receive far more hours of sunshine every day all year
- (D) A stream of cold air from space pours only onto the tropics

49. On a clear day a town's temperature rises after dawn, peaks in the early afternoon, then falls to its lowest just before the next dawn. This daily rise and fall is direct evidence that:

- (A) The Sun is the main heat source: temperature climbs while it warms the surface and falls once that heating stops
- (B) The barometer controls the temperature throughout the day
- (C) Humidity is what makes the Sun rise and set each day
- (D) Wind speed decides whether it is day or night

50. Why does the lowest temperature of a clear day usually come just before sunrise rather than at midnight?

- (A) The ground keeps losing heat all night, so it is coolest after the longest stretch without the Sun, just before dawn
- (B) The Sun secretly heats the ground at midnight and stops before dawn

- (C) The barometer pushes the temperature up again after midnight
- (D) Humidity always peaks at midnight and warms the air until dawn

51. Before instruments existed, people watched ants carry their eggs uphill or birds fly low to sense coming rain. What is the chief WEAKNESS of relying only on such nature's clues?

- (A) They give precise numbers for pressure and humidity
- (B) They are always perfectly accurate and never wrong
- (C) They are tied to one spot and give only vague, short-range hints rather than exact, comparable measurements
- (D) They gather data from across the whole country at once

52. How does meteorology improve on simply reading nature's clues?

- (A) It relies on watching a single animal as its only guide
- (B) It uses instruments to measure each element precisely and pools records from many places over time, so it can forecast ahead
- (C) It refuses to use any instruments and studies the sky only by eye
- (D) It can describe only today and can never predict the future

53. An Automated Weather Station (AWS) is set up on a remote Himalayan ridge. Why is an AWS especially suited to such a place compared with an ordinary station?

- (A) It needs an operator to read every instrument each hour by hand
- (B) Its sensors record and store data by themselves, so it keeps working where it is hard or dangerous for a person to stay
- (C) It works only inside large cities during the daytime
- (D) It measures the region's thirty-year climate instead of its weather

54. A school report on a recent cyclone must credit the official Indian body that gathers weather data nationwide and issues forecasts and storm warnings. The correct body is:

- (A) The National Rain Gauge Office (B) The Department of Climate History
- (C) The India Meteorological Department (D) The Indian Wind Vane Society

55. A forecast warns fishermen of an approaching cyclone so they stay ashore, and tells a city to clear its drains before heavy rain. This shows the chief value of forecasts is that they:

- (A) Stop the cyclone or the flood from ever forming
- (B) Give people and authorities time to prepare and stay safe before severe weather arrives
- (C) Make the weather change less often than it used to
- (D) Remove the need for any weather instruments

56. A weather log for one afternoon reads: 1 p.m. pressure 1012 mb, humidity 48 percent, light wind, clear sky; 2 p.m. pressure 1004 mb, humidity 67 percent, freshening wind, cloud building; 3 p.m. pressure 995 mb, humidity 88 percent, strong wind, dark cloud. The best forecast for the evening is:

- (A) Clear, calm and dry weather will continue (B) A cold, cloudless evening with no wind
(C) No change at all from the 1 p.m. conditions (D) Wet, stormy weather is approaching

57. In that same log the pressure falls 1012, then 1004, then 995 mb. Why is this steady fall an early warning, and what does it most likely mark?

- (A) Falling pressure often marks the approach of a depression, which is the seed of a storm
(B) Falling pressure marks the arrival of clear, settled fair weather
(C) Falling pressure means the station is steadily rising higher above sea level
(D) Falling pressure marks the complete end of all wind in the area

58. A station's readings are jumbled across a row: 1007, 33, 19, 58, belonging to pressure (mb), maximum temperature, minimum temperature and humidity (percent). The only sensible assignment is:

- (A) 1007 is the humidity, 33 is the pressure, 19 is the maximum temperature and 58 is the minimum temperature
(B) 1007 is the pressure, 33 is the maximum temperature, 19 is the minimum temperature and 58 is the humidity
(C) 1007 is the pressure, 19 is the maximum temperature, 33 is the minimum temperature and 58 is the humidity
(D) 1007 is the maximum temperature, 58 is the pressure, 33 is the humidity and 19 is the minimum temperature

59. A travelling pair compare notes from the same hour: traveller 1 in a valley logs 1006 mb and a maximum of 31 degrees; traveller 2 on a high ridge logs 905 mb and a maximum of 19 degrees. Which conclusion is best supported?

- (A) Traveller 2 is higher up, where the lower pressure and generally cooler air both fit a greater altitude
(B) Traveller 2 is lower down, because the smaller pressure number means lower ground
(C) Both travellers are at the same height, since pressure is the same everywhere
(D) Traveller 1 is higher up, because the larger pressure number means greater height

60. Which single statement chains together the chapter's ideas with EVERY part correct?

- (A) Pressure rises with altitude, is read by a thermometer, and a fall means clear skies
(B) Humidity is read by an anemometer in degrees, and 100 percent means very dry air
(C) Pressure falls with altitude, is read by a barometer in millibars, and a sharp fall can warn of a storm
(D) Rainfall is read by a wind vane in percent, and high humidity always brings rain

Solutions — Social Science (Class 7), Chapter 2: Understanding the Weather — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	D	21	D	31	D	41	A	51	C
2	C	12	A	22	B	32	A	42	A	52	B
3	A	13	A	23	A	33	D	43	C	53	B
4	B	14	D	24	D	34	A	44	C	54	C
5	B	15	A	25	C	35	C	45	D	55	B
6	C	16	B	26	A	36	C	46	D	56	D
7	B	17	B	27	B	37	D	47	C	57	A
8	D	18	D	28	C	38	C	48	A	58	B
9	B	19	C	29	D	39	B	49	A	59	A
10	D	20	C	30	D	40	B	50	A	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Weather is the condition of the atmosphere at a particular place and time; the dated entry with its noon drizzle is exactly that. Climate is the average of the weather taken over many years, which the 'decade after decade' paragraph describes. Having an exact number does not make a record climate, nor does the absence of one make it weather; and a record cannot be both at once just because it names the same town.

2. (C) Climate is the long-run average of weather over many years, so a single fine afternoon cannot redraw it; that afternoon is merely one day's weather. A wet climate can certainly have an occasional dry day, and weather and climate are distinct ideas, not synonyms.

3. (A) Latitude and day-length are not the only things that mould a climate; places sharing a latitude can still have their own long-term patterns. Both towns sit at essentially the same distance from the Sun, and the number of instruments cannot influence the actual climate.

4. (B) An immediate task hinges on the present state of the air, which is the weather; a choice spanning two decades depends on the area's usual long-term pattern, which is the climate. The two decisions therefore call on different scales of information, not the same one.

5. (B) The pattern across most years on record is the climate; a single unusual day is weather and cannot overturn a many-year average. The two halves of the report describe different things, one long-term and one momentary, so they are not both climate nor both weather.

6. (C) A climate label describes what is usual over many years, not a rule for every single day; individual days of weather can depart from the average, so a desert can still see a cold spell or a shower. Deserts do have dry climates, and weather and climate remain distinct concepts.

7. (B) The troposphere is the layer in contact with the ground, so the surface heated by the Sun, the water vapour rising from land and sea, and the rising warm air all meet there to form clouds, rain and storms. It is not the highest or thinnest layer, it is not in empty space, and oceans store water at the surface, not within a layer of air.

8. (D) Warm air expands and rises, lifting the top of the troposphere over the tropics; cold air contracts, so the layer sits lower over the poles. Cold air contracts rather than expands, all parts of Earth are roughly the same distance from the Sun, and rainfall does not set a layer's thickness.

9. (B) Air is a mixture of gases that does have weight, and the weight of the air column pressing down is atmospheric pressure. The atmosphere is neither rock nor purely water, and saying it is weightless yet somehow drives wind contradicts the idea of pressure that makes wind blow.

10. (D) Higher up there is less air stacked above, so the pressure drops and the thinner air carries less oxygen in each breath, which is why climbers feel breathless. Pressure falls with height rather than rising, it does not stay fixed, and it does not suddenly reach zero on a mountain pass.

11. (D) Pressure falls as one climbs, so the station with the highest pressure (1010 mb, P) is lowest and the one with the lowest pressure (700 mb, R) is highest, with Q in between. The largest number marks the lowest place, not the highest, the change with height is steady rather than random, and differing readings show the stations are at different heights.

12. (A) The barometer does read lower at the top, and the cause is exactly that less air lies above the summit to press down, so the reason properly explains the assertion. Both statements are true and they are linked as cause and effect.

13. (A) Temperature, atmospheric pressure, humidity and precipitation are all conditions of the air, so all four are weather elements. Each of the other sets smuggles in a land or water feature (rivers, mountains or soil type), which is not a condition of the atmosphere.

14. (D) Sea-level pressure sits near 1013 mb, so 1009 is the pressure; an everyday temperature is around 28 degrees; and humidity, a percentage, cannot exceed 100, so 64 percent fits. Reading 1009 as a temperature or a humidity is impossible, and swapping the 28 and 64 between temperature and humidity gives an unreasonable 64-degree reading.

15. (A) Weather elements describe the changing condition of the atmosphere; the time of sunrise is fixed by the regular motion of the Earth and Sun, not by the air, so it does not belong. It is not a kind of precipitation, and a barometer measures pressure, not sunrise.

16. (B) Words like 'warm' mean different things to different people, while a number on a fixed scale carries the same meaning for everyone, letting readings from many places be compared and combined. Numbers do not slow the weather, words are not banned, and people read instrument values all the time.

17. (B) Warming makes the liquid expand, and since the bulb is full the extra volume can go only up the thin tube, so the reading rises. The glass does not meaningfully widen, no air is pushed into a sealed tube, and the liquid does not float out.

18. (D) Celsius and Fahrenheit are simply two scales for reporting the same temperature, so the same warmth shows up as different numbers, and nothing is wrong; in fact 20 degrees Celsius equals 68 degrees Fahrenheit. The larger number is not automatically hotter on a different scale, Fahrenheit measures temperature not pressure, and one temperature legitimately gives different readings on different scales.

19. (C) The range is the high minus the low, 39 minus 21, which is 18 degrees; the mean is the two added and halved, (39 plus 21) divided by 2, which is 30 degrees. Adding instead of subtracting for the range gives 60, and confusing the two figures swaps 18 and 30.

20. (C) The ranges are W 6, X 27, Y 13 and Z 6, so Town X swings the most. A high minimum, a high maximum or a low maximum on its own does not decide the range, which is always the high minus the low.

21. (D) The ranges are W 6, X 27, Y 13 and Z 6, so W and Z share the smallest range of 6 degrees and are equally steady. The largest numbers, the lowest minimum, or the highest minimum do not by themselves mean the smallest range, which is the high minus the low.

22. (B) A bigger range means a bigger swing between the daytime high and the night-time low, so city Q's temperature varied much more even though both cities average 28 degrees. A matching mean with a wider swing does not make Q hotter all day, says nothing about rainfall, and two places certainly can share a mean while differing in range.

23. (A) M's mean is (30 plus 20) over 2, which is 25; N's mean is (26 plus 24) over 2, also 25, so the means match. M's range is 10 and N's is 2, so M swings more. A higher maximum does not make M warmer all day, N does not have a larger mean or range, and the ranges are clearly not equal.

24. (D) Precipitation covers every kind of water that falls from the sky, including rain, snow, sleet and hail. It is not limited to monsoon rain, it is not the invisible vapour still in the air, and it is certainly not air pressure.

25. (C) Hail is precipitation, hard ice that drops from the sky, whereas dew is moisture condensing onto cool surfaces such as grass without falling from a cloud. Dew does not fall as ice, neither is invisible vapour or wind, and neither is measured in percent or millibars.

26. (A) A rain gauge reports the depth the rain would form on flat ground, so 30 mm is a depth, not a time or a volume. Two gauges a few steps apart should record almost the same amount, so a wildly different reading points to a blocked or faulty gauge, not a real tenfold change over one small yard.

27. (B) A rain gauge is scaled to read the depth of liquid water, so frozen snow must be melted first and the resulting water measured. Snow is much fluffier than its meltwater, so its solid depth is misleading; snow is still precipitation, so zero is wrong; and there is no fixed doubling rule.

28. (C) Rain over a single small garden should fall in nearly equal depth everywhere, so a reading that is half the other points to a blockage or bad placement rather than a real change. Rain does not halve with every metre, a gauge does not switch to measuring temperature, and depth reflects how much fell, not how long it rained.

29. (D) Precipitation is water that falls from the sky, while humidity is the water vapour that remains in the air, so humidity is not precipitation. Humidity is not falling ice, it is reported as a percentage rather than in millimetres, and it is not the weight of the air.

30. (D) Wind blows from high to low pressure, and a bigger pressure difference over the same distance drives a faster wind. G-H differ by 28 mb while J-K differ by only 6 mb, so the wind is stronger between G and H. A smaller difference gives weaker wind, wind speed does depend on the difference, and air moves from high toward low, not the reverse.

31. (D) Air moves from higher to lower pressure, so with M at 1018 mb and N at 1002 mb the wind blows from M to N. It does not run from low to high, it does not simply circle in place, and the flow between two regions is sideways, not purely upward.

32. (A) Air does move to even out pressure differences, so the reason is true; but it does so by flowing from high to low pressure, which makes the assertion's 'low toward high' false. Hence the assertion is false while the reason is true.

33. (D) A wind vane points to show where the wind comes from, and an anemometer's spinning cups measure how fast it blows. Swapping the two reverses their jobs, and a barometer, thermometer, rain gauge or hygrometer measures pressure, temperature, rainfall or humidity, not wind direction or speed.

34. (A) A wind sock hangs limp when the air is nearly still and streams out straight when the wind is strong, so a limp-then-straight change means the wind strengthened over the morning. A straight sock signals strong wind, not calm; the sock shows wind, not pressure; and it does not measure rainfall.

35. (C) Slowly turning cups mean a low wind speed, and the wind vane points to where the wind comes from, here the north-west, so the wind is gentle and from the north-west. A vane does not show speed, the cups do not show direction, and the slow cups indicate weak wind rather than high pressure.

36. (C) A depression shows up first as falling pressure, which then drives a strengthening wind, the rising moist air builds cloud, and finally the cloud yields rain. Beginning with rain or with wind reverses the chain, and putting cloud before the pressure fall misses the earliest warning sign.

37. (D) A barometer measures the pressing weight of the air, while a wind vane and an anemometer measure the direction and speed of moving air, so all three deal with air. The rain gauge alone measures fallen water, making it the odd one out.

38. (C) Humidity needs a hygrometer and rainfall needs a rain gauge, and the station has neither, so those two elements cannot be recorded. It does have a thermometer, barometer, wind vane and anemometer, so temperature, pressure, wind direction and wind speed are all covered.

39. (B) Humidity is the amount of water vapour the air holds, reported as a percentage of the most it could hold. The weight of the air in millibars is pressure, the depth of rain in millimetres is rainfall, and wind speed is a different element again.

40. (B) A high relative humidity such as 92 percent means the air is nearly full of vapour, so very damp, while 28 percent means it is far from full, so dry. The high reading is not dryness, 92 percent does not guarantee rain, the 28 is a percentage not a temperature, and two different percentages cannot mean equal vapour.

41. (A) Drier air, far below its limit, can take up much more moisture, so clothes dry faster on the low-humidity Monday. Damp air near its limit accepts little more, drying is about humidity rather than a squeezing pressure, and humidity clearly does affect how fast washing dries.

42. (A) Rain needs the vapour to condense, which requires the air to cool sufficiently; high humidity on its own does not guarantee that, so a 94 percent day can stay dry. The

hygrometer is not faulty, humidity and rain are related, and 94 percent means very damp air, not dry air.

43. (C) At the same temperature, lower humidity means the air is further from full and can absorb more vapour, so evaporation is easier in town A and harder in the damp town B. Town B is the moister, not drier, town; measurable humidity does not mean rain; and humidity is not a measure of which town is hotter, especially when their temperatures are equal.

44. (C) A barometer measures the pressing weight of the air (pressure) and a hygrometer measures the water vapour in it (humidity), so they measure different elements. They do not both measure pressure or both measure humidity, and neither measures rainfall or wind speed.

45. (D) Humidity is measured by a hygrometer, not a barometer, so that pairing is wrong. Temperature with a thermometer, rainfall with a rain gauge, and wind speed with an anemometer are all correctly matched.

46. (D) The Sun is the engine of weather: its uneven heating of land, sea and air sets up the temperature and pressure differences that drive winds and other changes. The Moon's gravity, the spinning of instruments, and the barometers are not causes of the weather; instruments only measure it.

47. (C) The sand heats faster than the pond, the warm air over it rises and lowers the pressure there, and air flows in from the higher-pressure pond, which is the breeze. The heating is uneven rather than equal, sunlight does not physically blow air sideways, and the Sun warms rather than freezes the sand.

48. (A) Near the tropics the Sun is high overhead, so its rays fall almost straight down and pack heat into a small area; at the poles the same rays arrive at a slant and spread over a larger area, giving less heat per patch. The tropics are not closer to the Sun, the poles do not get more sunshine hours overall, and no special cold stream targets the tropics.

49. (A) Temperature tracks the Sun: it rises while the Sun heats the ground through the morning, peaks soon after midday, and falls through the night once that heating stops, which shows the Sun is the main heat source. A barometer measures pressure, humidity does not move the Sun, and wind speed does not set day and night.

50. (A) With the Sun absent all night the ground keeps radiating heat away and never stops cooling, so the coldest moment is at the end of that long Sun-free stretch, just before dawn, not at midnight. The Sun does not heat at midnight, the barometer does not raise temperature, and humidity does not warm the air through the night.

51. (C) Nature's clues warn of weather only locally and only roughly, with no exact figures and little notice, which is their main weakness. They do not yield precise numbers, are not

always right, and certainly do not pool data from across the country.

52. (B) Meteorology is the scientific study of weather: it measures each element with instruments, gathers data from many places over time, and uses that to forecast ahead, which nature's clues cannot do reliably. It does not depend on a single animal, it does use instruments, and its whole point is to predict, not just describe today.

53. (B) An AWS uses sensors that record data automatically with no person present, so it can keep gathering readings in remote or dangerous places. It does not need an hourly human operator, is not limited to daytime city use, and measures weather, not a thirty-year climate average.

54. (C) The India Meteorological Department is the official body that collects weather data across India and issues forecasts and warnings, including cyclone alerts. The other names are not the recognised national weather authority.

55. (B) Forecasts give advance warning so that people and authorities can prepare and reduce harm, as with fishermen staying ashore and drains being cleared. A forecast cannot prevent a storm forming, does not slow how often weather changes, and depends on instruments rather than replacing them.

56. (D) Falling pressure, rising humidity, strengthening wind and thickening dark cloud together are the classic signs of an approaching storm, so wet, stormy weather is on the way. Clear, calm, cloudless or unchanged conditions all contradict the worsening trend in every reading.

57. (A) A depression is a region of low pressure that can grow into a storm, so a steadily falling barometer is an early warning of an approaching depression. Falling pressure does not signal fair weather, a fixed station does not change altitude, and the log itself shows the wind strengthening, not ending.

58. (B) Pressure sits near 1013 mb, so 1007 is the pressure; humidity is a percentage at or below 100, so 58 is the humidity; and of the two temperatures the larger, 33, is the maximum and the smaller, 19, is the minimum. Calling 1007 a humidity or a temperature is impossible, and the maximum cannot be smaller than the minimum.

59. (A) Pressure falls and the air tends to be cooler with height, so traveller 2's lower pressure of 905 mb and cooler reading point to the higher position on the ridge. A smaller pressure number marks a higher place, not a lower one; pressure is not the same everywhere; and the larger pressure number belongs to the lower valley.

60. (C) Pressure does fall as one climbs, a barometer reads it in millibars, and a sharp drop warns of a possible storm, so every part of that statement is right. The others fail somewhere: pressure falls rather than rising and is read by a barometer not a thermometer; humidity is read by a hygrometer in percent and 100 percent means

saturated, not dry; and rainfall is read by a rain gauge in millimetres while high humidity does not guarantee rain.